Bakeware: HMTc Standard

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CORRESPONDENCE

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for bakeware in Food-Contact Consumer Goods and Kitchenware, evaluated on element RELEASE FROM THE FOOD-CONTACT ALLOY INTO FOOD OR A FOOD SIMULANT, µg/kg of food, determined against an acidic simulant at the article's stated use temperature and duration with the third successive release governing for repeat-use articles. This is a RELEASE basis on the food, not the plastics-simulant convention and not a total-content basis: the percent-by-weight lead limits on food-contact metals and solder are a separate MATERIAL-CONTENT obligation, disclosed as companions and never published here.. No government sets a numeric heavy-metal limit for this product form, so every panel analyte is held under mandatory analytical control or a substantiated not-material determination rather than an invented ceiling.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum

Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Lead in a food-contact metal article arrives from the alloy, from lead-bearing solder and brazing at joints and

Cadmium (Cd)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Cadmium in a food-contact metal article arrives from cadmium-bearing brazing fillers and from cadmium plating. The EU restricts both by content under REACH Annex XVII (Reg (EU) 494/2011), which is a percent-by-weight restriction and not a release ceiling; it is verified as a separate content obligation. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Total arsenic (tAs)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Arsenic is a residual impurity of copper and lead alloys and of some brasses and bronzes. It is determined as the total element; no food-contact instrument in the credible set states arsenic as a species. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Inorganic arsenic (iAs)	control	Arsenic released from a metal or alloy surface is inorganic by construction, so an inorganic-arsenic determination cannot be lower than the total-arsenic result by any meaningful margin and cannot be bounded by a published total-arsenic number, because this row's total-arsenic cell is itself a control with no sovereign ceiling. A not-material determination is therefore not available here and the fail-safe applies: inorganic arsenic is a control, tested and escalated alongside total arsenic. No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one.

Mercury, total (tHg)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Mercury has no functional role in food-contact alloys and appears only as a trace impurity, but it is a mandatory panel cell and is determined as the total element on every food-contact component. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic food chains, where an anaerobic sulfate- or iron-reducing microbial community acts on a bioavailable inorganic-mercury pool over long residence times. This row's THREE named material configurations are each addressed rather than covered by a single clause. An aluminised, tinned, aluminium or carbon-steel pan is a metal surface with no organic matrix and no methylation pathway. A moulded SILICONE bakeware article and a non-stick or silicone-release COATING are cured synthetic polymers: they are organic, but they are crosslinked solids with no bioavailable carbon substrate for a methylating community, they are used in dry, aerobic, high-temperature baking contact rather than in a submerged anoxic environment, and they carry no inorganic-mercury pool to methylate. A GLASS or CERAMIC baking dish is a fired inorganic matrix with no organic carbon at all. None of the three supplies a methyl donor to mercury, and the baking contact conditions this row certifies — high temperature, short duration, washed and dried between uses — are the opposite of the conditions methylation requires. No credible sovereign sets a methylmercury requirement for any food-contact material in any basis. Total mercury is the governing determination for this row and remains a mandatory control on every configuration; methylmercury is not material and never substitutes for it.

Nickel (Ni)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Nickel release from austenitic stainless steel and from nickel-plated surfaces into acidic and saline food is a real, documented and material pathway, and it is exactly the pathway for which no sovereign ceiling exists. The EU plastics nickel SML of 0.02 mg/kg food is not applied here: it was written for a polymer surface and a bare stainless-steel surface does not meet it, so importing it would publish an unfounded number rather than a protective one. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Tin (Sn)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Tin release arrives from tinned linings, tinplate and tin-bearing solder. The sovereign tin numbers that exist are maximum levels on canned food, a FOOD basis on the contents rather than the article, and they are not published as this row's release ceiling. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Aluminium (Al)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Aluminium release from bare and anodised aluminium into acidic, saline and long-contact food is a real and documented pathway, and no sovereign sets a release ceiling for it. The EU plastics aluminium SML of 1 mg/kg food is not applied to an aluminium article: the instrument was written for plastics and did not contemplate an article whose food-contact surface is the element itself. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Total chromium (Cr)	control	No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one. Total chromium is retained as a screening determination for the alloy-release pathway rather than as a toxicological ceiling: chromium released from austenitic stainless steel is overwhelmingly chromium(III), which is the benign species, so a total-chromium number on this row would not be protective even if a sovereign had set one. The health-relevant determination is the hexavalent-chromium cell. Bakeware release testing must be run at the stated baking temperature and duration rather than only at the ambient acidic-simulant convention, because dry high-temperature contact is this row's actual use condition.

Hexavalent chromium (Cr-VI)	control	Hexavalent chromium is the health-relevant chromium species and it cannot be dismissed on a metal article: chromate conversion coatings, passivation treatments and chromate-bearing pigments in interior printing are live sources, and hot alkaline conditions can oxidise chromium(III) at the surface. No credible sovereign sets a hexavalent-chromium release ceiling for a food-contact metal article, and this row's total-chromium cell is itself a control, so no published parent bounds this species and a speciation-bounded not-material determination is unavailable. The fail-safe applies. No credible sovereign sets a numeric ceiling on element release from a metal or alloy food-contact article. Regulation (EU) 10/2011 governs plastics only; metals and alloys are a separate Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, the US regulates food-contact substances without an element release ceiling, and Codex, FSANZ, Health Canada, the UK and Hong Kong set none. Testing is therefore mandatory on every certified lot with no invented ceiling: a limit borrowed from the plastics regulation would be a limit written for a different material family, and for an alloy surface the plastics nickel and aluminium values are not even achievable, so borrowing them would be an unfounded number rather than a stricter one.

4 | Cite this standard

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/bakeware. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

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